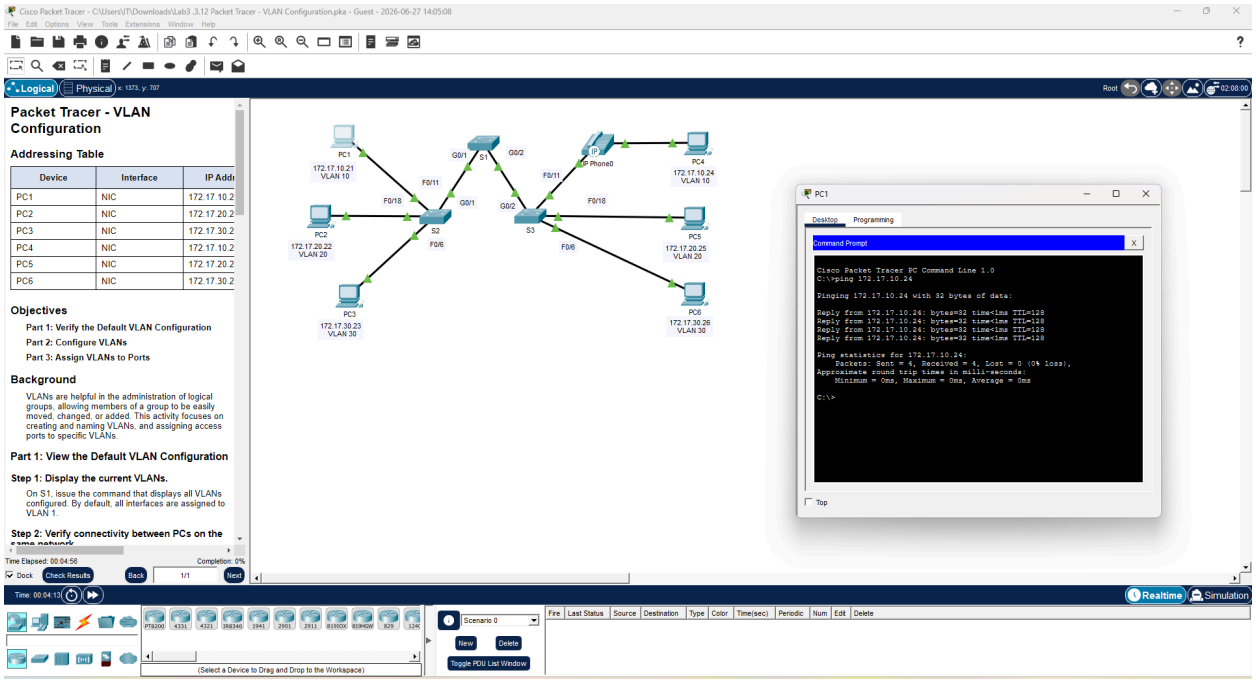
Lab 3: VLAN Configuration Report

1. Objectives



The primary goals of this lab were to:

- Verify the default VLAN configuration on Cisco switches.
- Create and name multiple VLANs (Faculty/Staff, Students, Guest, Management, and VOICE).
- Assign specific access ports to their respective VLANs.
- Configure Voice VLAN and Quality of Service (QoS) on applicable interfaces.

2. Network Topology & Architecture

The network infrastructure utilizes a hierarchical design comprising three switches:

- **Switch 1 (S1):** Functions as the Core/Distribution switch, providing central aggregation for S2 and S3.
- **Switch 2 (S2):** An access switch hosting PC1 (VLAN 10), PC2 (VLAN 20), and PC3 (VLAN 30).
- **Switch 3 (S3):** An access switch hosting PC4 (VLAN 10), PC5 (VLAN 20), PC6 (VLAN 30), and a Cisco IP Phone (VLAN 150).

3. Master Addressing Table

Device	Interface	IP Address	Subnet Mask	VLAN
PC1	NIC	172.17.10.21	255.255.255.0	10 (Faculty/St...
PC2	NIC	172.17.20.22	255.255.255.0	20 (Students)
PC3	NIC	172.17.30.23	255.255.255.0	30 (Guest)
PC4	NIC	172.17.10.24	255.255.255.0	10 (Faculty/St...
PC5	NIC	172.17.20.25	255.255.255.0	20 (Students)
PC6	NIC	172.17.30.26	255.255.255.0	30 (Guest)

4. Device Configurations

4.1 Switch 1 (S1) Configuration

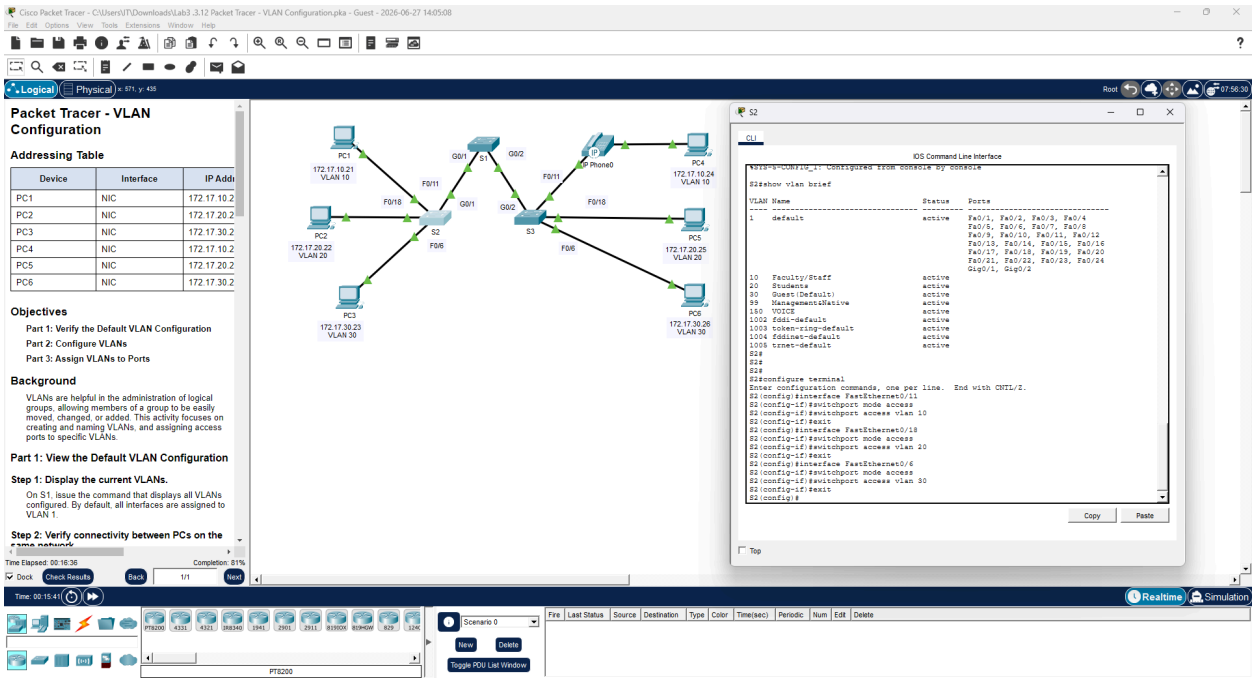
S1 acts as the database master for all required VLANs.

```
None
S1> enable
S1# configure terminal
S1(config)# vlan 10
```

```
S1(config-vlan)# name Faculty/Staff
S1(config)# vlan 20
S1(config-vlan)# name Students
S1(config)# vlan 30
S1(config-vlan)# name Guest(Default)
S1(config)# vlan 99
S1(config-vlan)# name Management&Native
S1(config)# vlan 150
S1(config-vlan)# name VOICE
S1(config-vlan)# exit
```

4.2 Switch 2 (S2) Configuration

S2 defines local VLANs and assigns them to specific host-facing interfaces.



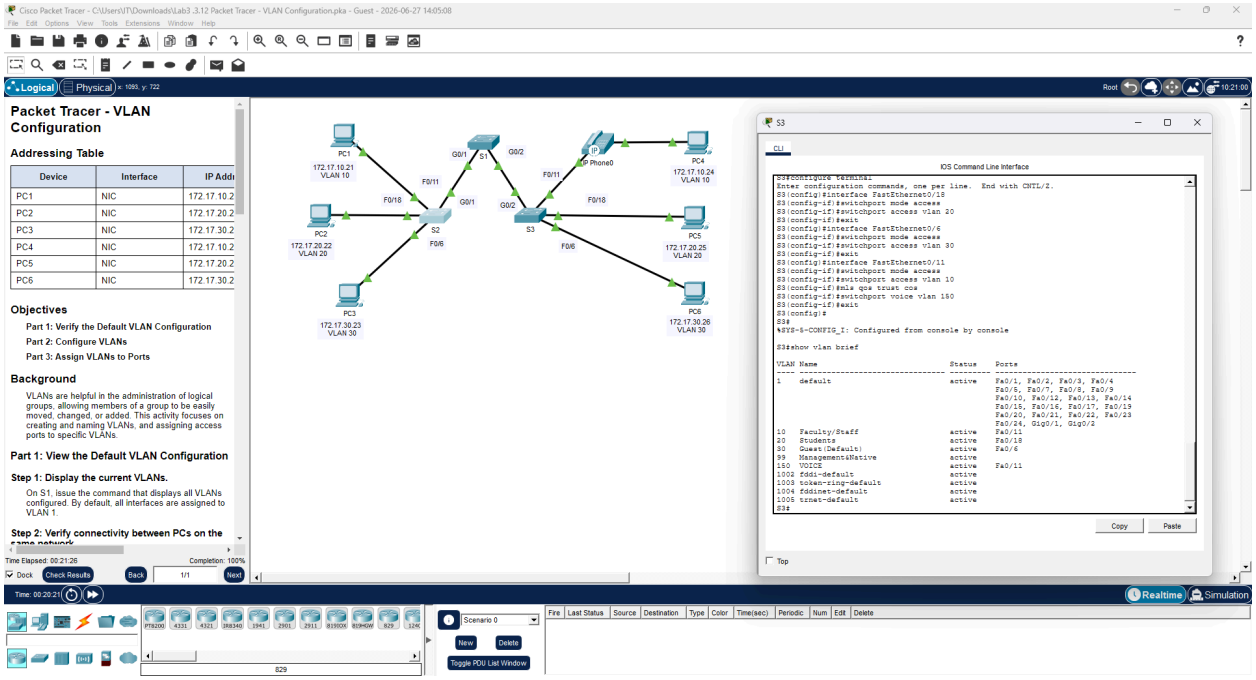
None

```
S2(config)# interface FastEthernet0/11
S2(config-if)# switchport mode access
S2(config-if)# switchport access vlan 10
S2(config)# interface FastEthernet0/18
S2(config-if)# switchport mode access
S2(config-if)# switchport access vlan 20
```

```
S2(config)# interface FastEthernet0/6
S2(config-if)# switchport mode access
S2(config-if)# switchport access vlan 30
```

4.3 Switch 3 (S3) Configuration

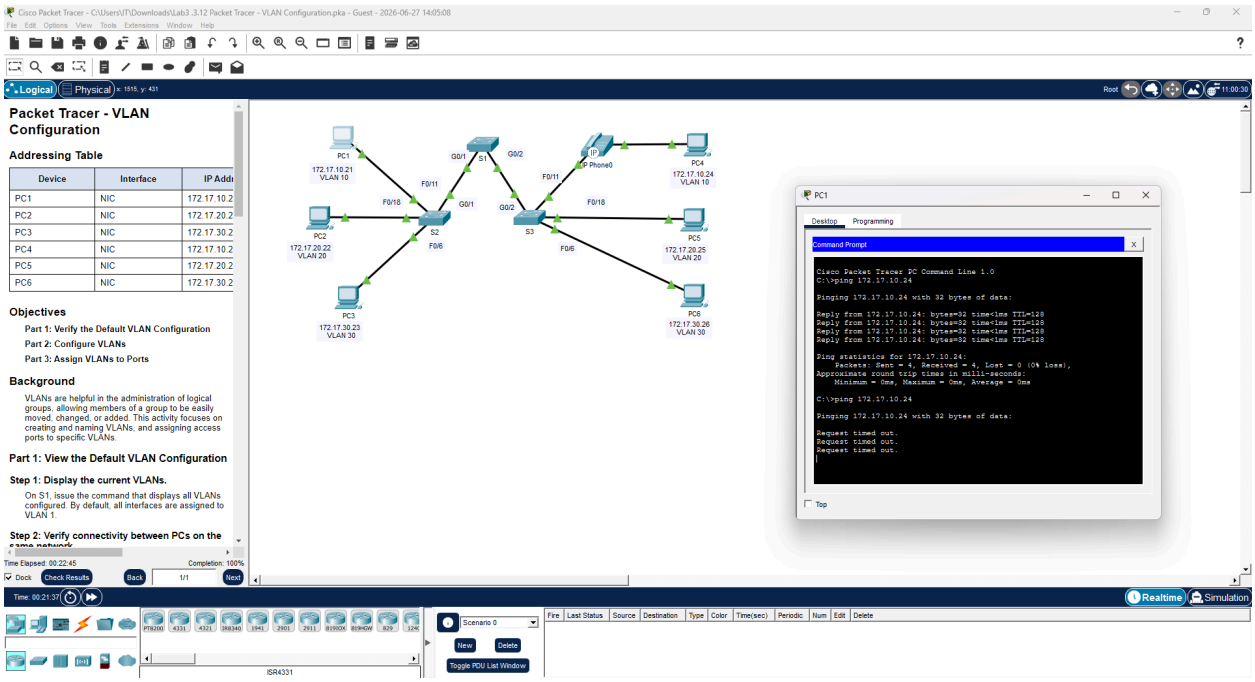
S3 includes Voice VLAN integration and QoS trust settings for the IP phone interface.



None

```
S3(config)# interface FastEthernet0/18
S3(config-if)# switchport mode access
S3(config-if)# switchport access vlan 20
S3(config)# interface FastEthernet0/6
S3(config-if)# switchport mode access
S3(config-if)# switchport access vlan 30
S3(config)# interface FastEthernet0/11
S3(config-if)# switchport mode access
S3(config-if)# switchport access vlan 10
S3(config-if)# mls qos trust cos
S3(config-if)# switchport voice vlan 150
```

5. Lab Questions & Answers



Q1: What benefits can VLANs provide to the network?

VLANs provide improved security by isolating sensitive data, cost reduction through more efficient use of existing bandwidth and links, and better performance by reducing the size of broadcast domains. They also offer simplified administration when moving or adding users to logical groups.

Q2: Which command will only display the VLAN name, status, and associated ports on a switch?

The command is show vlan brief.

Q3: Although the access ports are assigned to the appropriate VLANs, were the pings successful? Explain.

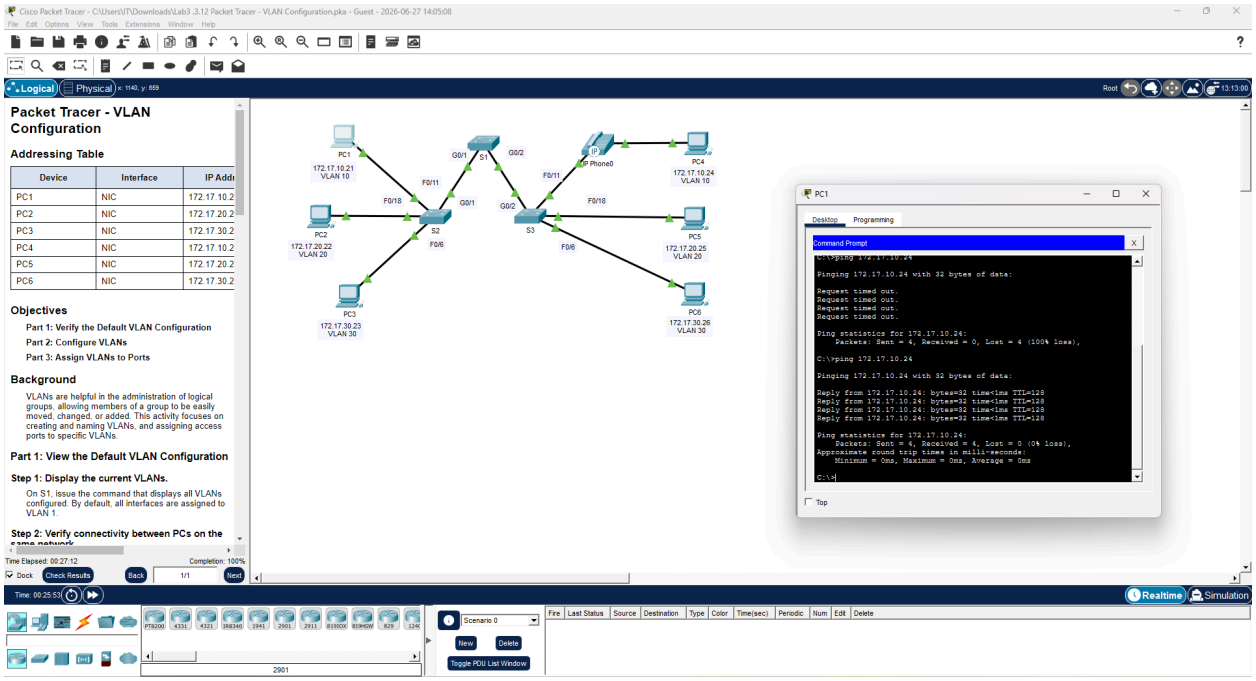
No, the pings were not successful. While the host ports were correctly assigned to their VLANs, the inter-switch links (Gig0/1 and Gig0/2) were still operating as access ports in the default VLAN 1.

Q4: What could be done to resolve this issue?

The interconnected ports between switches must be configured as Trunk ports using IEEE

802.1Q encapsulation. This allows the switches to pass traffic for all VLANs across the shared physical links.

6. Conclusion & Reflections



This lab demonstrated that physical connectivity and matching IP subnets are insufficient for communication once Layer 2 segmentation (VLANs) is introduced. The successful integration of Voice and Data on S3 interface F0/11 highlights how modern networks maximize cabling infrastructure while ensuring traffic quality via QoS. Proper trunking configuration remains the critical final step for multi-switch VLAN environments.